

APPLIED SCIENCES AND ENGINEERING

Eliminating performance loss from perovskite films to solar cells

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Preoptimizing perovskite films may generally improve the performance of the final perovskite solar cells (PSCs). However, the research on whether the film optimization fully contributes to the enhancement of the final PSCs has been long neglected. We demonstrated that the preparation of metal electrodes by high-vacuum thermal evaporation, an unavoidable step in almost all device fabrication processes, will damage the surface of perovskite films, resulting in component escape, defect density rebound, carrier extraction barrier, and film stability deterioration. Therefore, the prepared perovskite film and the final film actually working in devices are not exactly the same, and the contribution of film optimization to the device improvement was weakened. We designed a bilayer structure composed of graphene oxide and graphite flakes to eliminate the unwanted film inconsistencies and thus save the film optimization loss. Therefore, the efficient PSCs with power conversion efficiency of 25.55% were obtained, which demonstrated negligible photovoltaic performance loss after operating for 2000 hours.

INTRODUCTION

As the latest generation of photovoltaic technology, perovskite solar cells (PSCs) are explosively attracting attention from academia and industry (1–5). Although solar cell device is a complex system composed of multiple functional layers (6), optimizing the perovskite film could generally contribute to the enhancement of final performance of PSCs (7–10). Previously, the preprepared or optimized perovskite films were assumed to be the same as the films actually working in the device (11–13). However, whether there is inconsistency between them has been long neglected and is well worth investigating to maximize the contribution of film optimization to the PSC performance improvement, especially at this critical moment when the efficiency of PSCs is approaching that of silicon solar cells (14).

Because perovskite is a newly emerging semiconductor material, the main preparation and characterization methods for both materials and devices are borrowed from conventional semiconductors. However, perovskite has markedly different crystal properties, so whether these borrowed methods are still perfectly applicable to perovskite needs to be carefully considered, such as the preparation of metal electrodes by thermal evaporation under ultrahigh vacuum. Perovskite has a markedly weak chemical bonds and soft lattice compared with those conventional semiconductor materials (15), and the 12 orders of magnitude higher vacuum (from 10^5 to 10^{-7} Pa) coupled with heat may damage the perovskite film, resulting in the film actually working in the device being worse

than the preprepared or optimized film (15–24). Considering that the PSCs prepared by nonthermal evaporative electrodes (printing or hot pressing) exhibit drastically better stability (such as up to 5000 and 9000 hours) (25–28) inspired us to further investigate whether the high-vacuum thermal evaporation electrode preparation (HTEE) itself has a negative effect on the perovskite film and whether there is a discrepancy in perovskite film before and after HTEE. Changes in the perovskite film due to HTEE were often not considered previously.

Herein, we demonstrated that during HTEE, the coupling effect of 12 orders of magnitude elevated vacuum and heat will degrade the surface of perovskite film by facilitating the escape of formamidinium (FA), methylammonium (MA), and iodide (I) ions. This previously default harmless step was found to cause interface nonradiative recombination, weaken the ultrafast carrier extraction, and worsen the film stability. Even if the surface was passivated by post-fabricated two-dimensional (2D) layers, HTEE could still trigger defect density rebound. We engineered a bilayer structure of highly conductive, chemically inert graphene oxide (GO) and graphite flakes (GOG) in the device to effectively eliminate the inconsistencies between the prepared perovskite film and the film actually working in PSCs, thus saving the film optimization loss. Moreover, extensive delicately designed experiments also confirmed that the multifunctional GOG could markedly improve the moisture, light, and thermal stability of perovskite films by inhibiting ion migration and component escape.

RESULTS

Damage of perovskite film by HTEE

The (001) diffraction peak of the perovskite film (FAMAPbI_3) was first found to lightly shift by 0.04° after HTEE at 10^{-7} Pa (fig. S1), which was reaffirmed by constructing two samples from the same perovskite film, except that one of them has undergone HTEE (note S1), implying that perovskite films have been affected by HTEE. After preparing the electrode, we removed the electrode and Spiro-OMeTAD to obtain the bare perovskite film that has undergone HTEE for subsequent film characterization, and the process of removing these two layers was

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proved to not affect the underneath perovskite film (note S2). Although the morphology of the perovskite films did not change markedly after HTEE (fig. S5), the x-ray photoelectron spectroscopy (XPS) characterization with detection depth of several nanometers revealed that HTEE weakened the FA/MA intensity on the surface of perovskite film (Fig. 1, A and E). This could also be observed in a wider region (500 μm by 500 μm) through surface XPS mapping on N 1s orbits of FA/MA at 400.8 eV (Fig. 1, D and H) (29). More unexpectedly, we detected a very strong I 3d signal on the freshly prepared gold electrode (note S3), which could only be escaped from the perovskite film (fig. S7), solidly confirming the surface components loss of perovskite film during HTEE. As expected, the I/Pb ratio on the perovskite surface decreased from 3.1 to 2.7 after HTEE (fig. S8). Further, we prepared two exactly equal amounts of perovskite powder scraped from fresh perovskite films, one of which subsequently undergone HTEE environment, while another without. Then, an equal amount of EDTA as the internal standard was added for hydrogen nuclear magnetic resonance (^1H -NMR) spectroscopy characterization (fig. S9) (30). The integral peak area ratio of FA to internal standard showed that the HTEE process markedly decomposed the perovskite component (Fig. 1C and fig. S9C) (31). It should be noted that the powder has a higher specific surface area to be exposed to the HTEE environment and therefore exhibited more drastic component loss.

Atomic force microscopy–based infrared (AFM-IR) spectroscopy was used to further investigate the organic component changes on the surface of perovskite films before and after HTEE (32). It can be found that the IR intensities recorded at 1466 cm^{-1} (C–N of MA⁺) and 1712 cm^{-1} (C=O of FA⁺) were weakened after HTEE (Fig. 1, I to L) (28, 30), consistent with the results of micro-IR spectroscopy characterization (fig. S10). Last, we directly compare the component distribution of the half-device without depositing the electrode and the complete device by time-of-flight secondary ion mass spectrometry (TOF-SIMS). From fig. S11, there was no difference in the distribution of other functional layers (Spiro-OMeTAD, SnO₂, and ITO) before and after HTEE. However, we found that the distribution of FA, MA, and I were markedly reduced on the surface of perovskite films after HTEE (Fig. 1, E to G) while remaining consistent in the bulk, strongly supporting the conclusion that HTEE degraded the surface of the perovskite film. Note that the characteristic signals of perovskite did not start from $t = 0$ s because of the presence of Spiro-OMeTAD+Au on the perovskite layer. The corresponding position when there is a clearly defined perovskite characteristic signal was considered as the surface of the perovskite film. In general, the ultrahigh vacuum coupled with heat during HTEE may damage the perovskite film surface by breaking the weak chemical bonds (20–22, 24), thus leading to the escape of the organic components

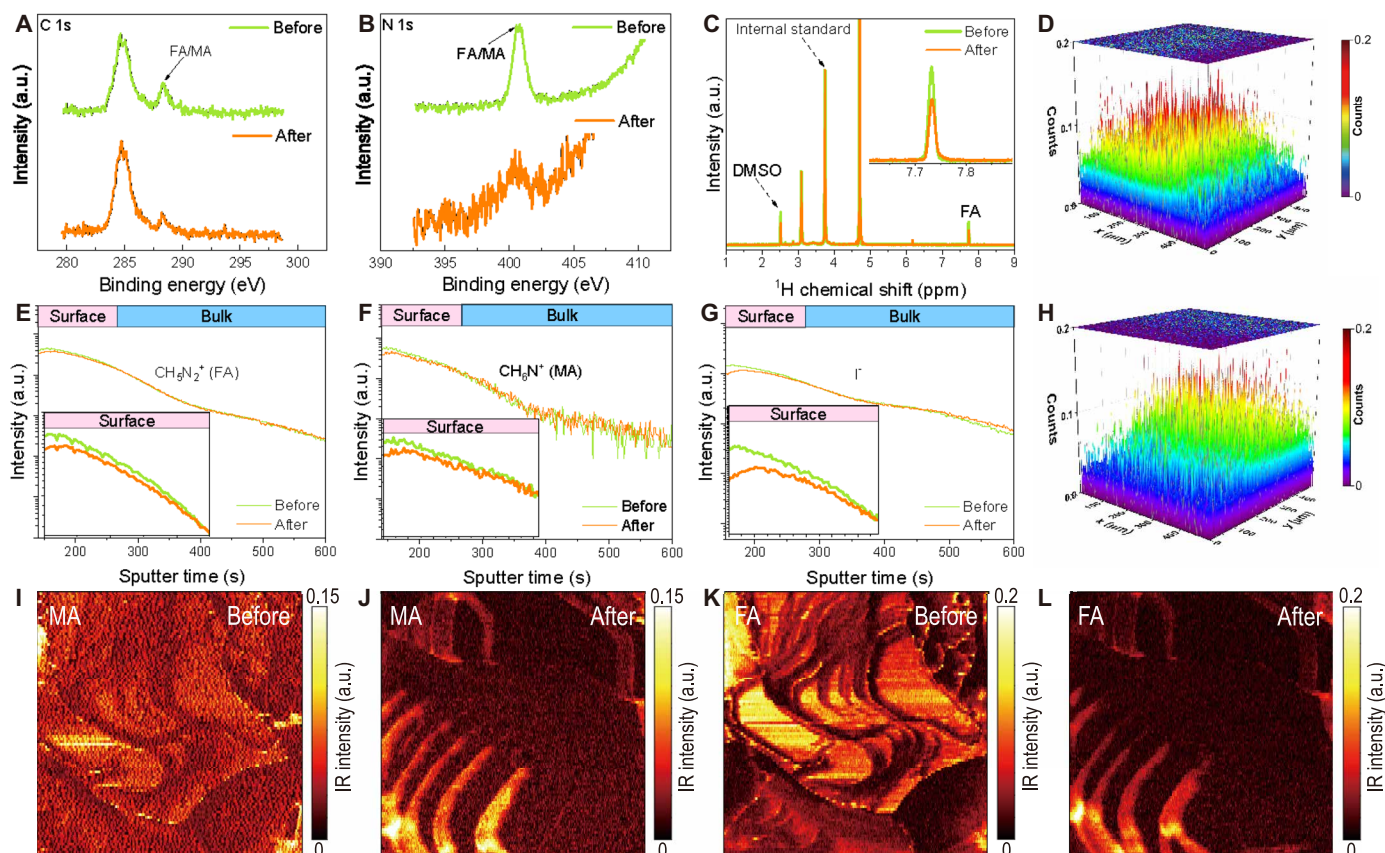


Fig. 1. HTEE damages the surface of the perovskite film. (A and B) XPS results of C 1s and N 1s orbits of perovskite films before and after HTEE. (C) ^1H -NMR of perovskite powder before and after HTEE. The inset is the amplified FA signal. (D and H) XPS mapping of N 1s orbits of perovskite films before (D) and after (H) HTEE. (E to G) TOF-SIMS results of FA, MA, and I distribution in half-device without depositing the electrode (ITO/SnO₂/perovskite/Spiro-OMeTAD, marked as “Before”) and the complete device (marked as “After”). The inset is the amplified distribution in the surface region. (I and J) AFM-IR images recorded at 1466 cm^{-1} for MA before and after HTEE; 1 μm by 1 μm . (K and L) AFM-IR images recorded at 1712 cm^{-1} for FA before and after HTEE; 1 μm by 1 μm .

and iodine at high negative pressure (fig. S12). The decomposition of perovskite is reversible (33), and the escape of components caused by vacuum extraction will, in turn, exacerbate the decomposition reaction. Therefore, we suspect that the photoelectric properties of the perovskite film will be affected inevitably.

Loss of optoelectronic properties

Conductive AFM (c-AFM) showed that HTEE weakened the conductivity of the perovskite film surface (Fig. 2, A and B), substantially reducing the maximum current from 1400 to 700 pA when a

voltage of 0.6 V was applied to the probe (Fig. 2C). Moreover, single-grain c-AFM results revealed that the electrical conductivity on the perovskite grain surface rather than the grain boundary was mainly reduced (Fig. 2, D and E). Steady-state photoluminescence (PL) and time-resolved PL (TRPL) with detection depth within 100 nm further demonstrated the increased defect density on the surface of perovskite films during the HTEE process, which reduced the PL intensity and lifetime from 2.4 to 1.7 μ s (Fig. 2F). TRPL mapping also definitely showed that HTEE weakened the PL lifetime of perovskite film (Fig. 2, G and H). Moreover, the lifetime distribution on the film

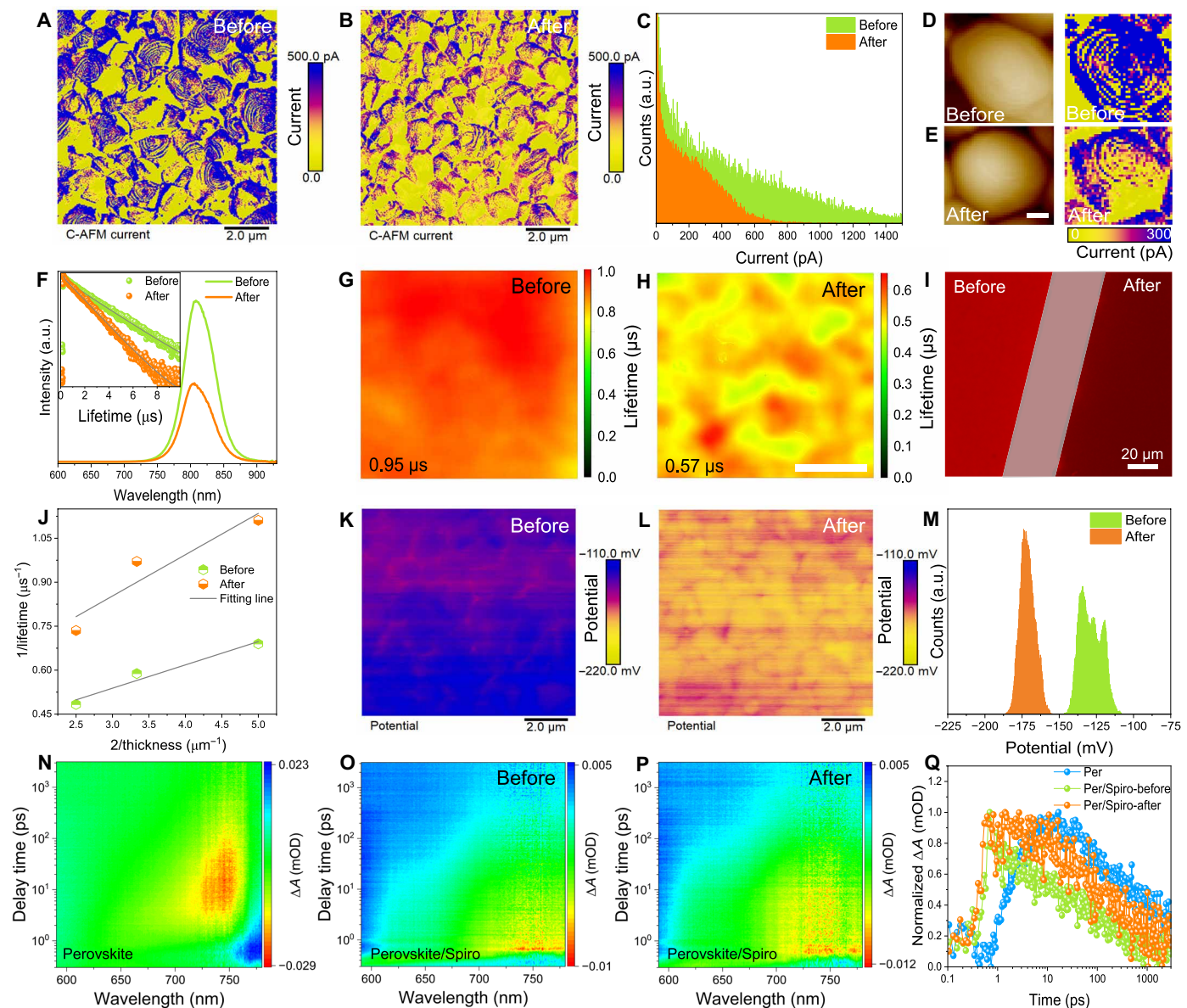


Fig. 2. Optoelectronic properties characterizations of perovskite film. (A) c-AFM image of perovskite film. (B) c-AFM image of perovskite film after HTEE. (C) Current comparison of perovskite film surface extracted from c-AFM image before and after HTEE. (D and E) c-AFM characterization of single perovskite grains before and after HTEE. Scale bars, 300 nm. (F) PL and TRPL results of perovskite film before and after HTEE. Insert is TRPL results. (G and H) TRPL mapping of perovskite film before and after HTEE. Scale bars, 10 μ m. (I) CLSM mapping of perovskite film before and after HTEE in one field of view. (J) Functional relationship between 2/thickness and 1/lifetime. (K and L) Contact potential difference of perovskite film surface before and after HTEE. (M) Comparison of contact potential difference on perovskite film surface before and after HTEE. (N) Pseudo color fs-TA mapping of bare perovskite film. (O) Pseudo color fs-TA mapping of perovskite/Spiro-OMeTAD film. (P) Pseudo color fs-TA mapping of perovskite/Spiro-OMeTAD film after HTEE. (Q) Bleach recovery kinetics probed at 760 nm of three samples.

surface after HTEE is not uniform (Fig. 2H), which was also validated by conducting TRPL mapping at other regions (fig. S13), suggesting that maybe the local film surface has been damaged. The weakened optoelectronic property was also reproved by absolute PL quantum yield, which was reduced from 3.9 to 2.4% (fig. S14F).

We further designed experiments to simultaneously observe the PL properties of perovskite films before and after HTEE in one field of view using a confocal laser scanning microscope (CLSM) (note S4). The CLSM mapping result more directly showed that the sample after HTEE has a lower PL intensity (Fig. 2I). The above pieces of evidence confirm that the surface of perovskite films has been damaged by HTEE; to further separate the effect of HTEE on surface and bulk, we conducted the thickness-dependent TRPL decay kinetics characterization (Fig. 2J and note S5) (34). The results showed that HTEE increased the surface recombination rate (S) of the perovskite film from 7.9 to 14 cm/s (fig. S16C), while it had relatively little effect on bulk recombination lifetime (τ_b), which further clarified that HTEE mainly deteriorated the surface of the perovskite film. There are two possible factors that affect the surface of perovskite film during the HTEE process, that is, high vacuum and heat (fig. S17A). We only subjected a sample to extremely high vacuum without applying heat (fig. S17, C and D), and the results demonstrated that only vacuum just contributed about 32% to total TRPL lifetime reduction, while the vacuum-coupled heat markedly exacerbated the damage to the perovskite film surface during HTEE, resulting in inconsistencies between the prepared perovskite films and the perovskite films actually working in the devices. Moreover, the effect of light on the HTEE process is negligible because of the presence of opaque Spiro-OMeTAD (fig. S17, E to H).

The interface energy level arrangement is crucial for the extraction of charge carriers. Kelvin probe force microscopy (KPFM) results revealed that HTEE caused a more negative surface potential of 45 mV (Fig. 2, K to M), which means that the Fermi level has shifted toward conduction band at the surface of perovskite film, resulting in a more n-type interface (14). Cross-sectional KPFM reconfirmed that the potential distribution in “after” perovskite film decreases on the surface (fig. S18), which is undesirable for carrier interface extraction. To verify it, femtosecond transient absorption spectroscopy (fs-TA) was carried out to detect ultrafast carrier interface extraction (35). As shown in Fig. 2 (N to P) and bleach recovery kinetics probed at 760 nm in Fig. 2Q, it can be found that bleach recovery for perovskite/Spiro-OMeTAD sample is much faster than that of bare perovskite film, indicating that the presence of Spiro-OMeTAD efficiently extracts photogenerated carriers on picosecond scale, thus accelerating the decay of carriers in the film (36–38). However, the “After” sample showed a longer decay time, implying that the rapid extraction of charge carriers was suppressed to some extent because of interface barriers (39). These results indicate that HTEE brings unwanted inconsistencies between the prepared perovskite film and the film actually working in the device by disrupting the interface energy levels. Last, we found that HTEE also slightly weakened the stability of the perovskite film (fig. S19), which is understandable because the surface of the film was damaged and therefore had lower resistance to external invasion.

Rebound of defect density

A consensus in the perovskite community is that passivating the upper surface of perovskite films is an extremely important prerequisite for constructing high-performance PSCs, so it is necessary to investigate

the effect of HTEE on passivated perovskite films. We selected two most commonly used passivators, 3-fluorophenylethylammonium iodide (3F-PEAI) and *n*-octylammonium iodide (OAI) to construct 2D passivation layers on perovskite film surface. TRPL mapping results (Fig. 3, A to H, and fig. S20) show that OAI and 3F-PEAI can indeed reduce the defect density on the perovskite film surface to improve the TRPL lifetime from 0.95 to 1.41 (Fig. 3B) and 1.18 μ s (Fig. 3F), respectively. However, after HTEE, their lifetimes were both weakened from 1.41 to 1.29 μ s for the OAI passivator (Fig. 3, C and D) and from 1.18 to 1.13 μ s for the 3F-PEAI passivator (Fig. 3, G and H), proving the rebound of surface defect density. The TRPL lifetime reduced by HTEE accounts for 26 and 22% of the total lifetime increased because of passivation of these two passivators, respectively (Fig. 3, D and H). Note that despite this, both the films after HTEE still showed much longer lifetime than that of the unpassivated films. That is why we can still substantially improve the photovoltaic performance of devices by prepassivating the film. Unfortunately, the film passivation was not completely contributed to the final device improvement because of the rebound of defect density caused by HTEE.

We further found that the diffraction peak of 2D passivation layer in OAI-passivated film was weakened after HTEE by small-angle x-ray diffraction (XRD) (Fig. 3I). PL results further show that the emission intensity of 2D phase with $n = 1$ (OA_2PbI_4) and 3D perovskite phase were obviously weakened after HTEE, while the PL intensity of 2D phase with $n = 2$ ($\text{OA}_2\text{FAPbI}_7$) was only slightly reduced (Fig. 3J). The synchrotron radiation-based grazing-incidence wide-angle x-ray scattering (GIWAXS) also further verified that $n = 1$ phase is more damaged than $n = 2$ phase after HTEE (Fig. 3, K to M). For the 3F-PEAI passivator, we also found that $n = 1$ phase is more damaged than $n = 2$ phase after HTEE (fig. S21B), whereas the difference with OAI passivator is that the weakening of 3D perovskites was relatively mitigated in the 3F-PEAI-passivated sample. fs-TA was used to further investigate this interesting finding. Although we failed to detect the 2D phase, it was still verified that the absorption intensity of 3D perovskite in OAI-passivated sample was decreased after HTEE (fig. S22). However, the absorption decrease of 3D perovskite in 3F-PEAI-passivated sample was alleviated (fig. S23). In the 2D passivation layer, $n = 1$ phase was proved to be distributed in the outermost (40); we therefore speculate that it is more susceptible to be damaged in both the OAI- and 3F-PEAI-passivated samples (fig. S24), while 3F-PEAI-induced 2D phase may be able to protect the underlying 3D perovskite phase to a certain extent. In general, although we can effectively improve the final device performance by prepassivating the perovskite film (41), because of the inconsistency between the optimized film and the film actually working in the device caused by the rebound of defect density after HTEE, we unfortunately lost part of the contribution of the film optimization to the device performance enhancement.

Eliminating the inconsistencies

Given that only the surface of the perovskite film was destroyed by HTEE, we therefore expect that inserting a thick protective layer may weaken the effects of HTEE on the perovskite film. The requirements for the insertion layers are very demanding: It must have a certain thickness and airtightness, high conductivity, chemical inertness, matching energy levels, easy preparation, and capability of inhibiting ion migration. After elaborately screening, we designed a bilayer structure consisting of GO and lying flat natural graphite

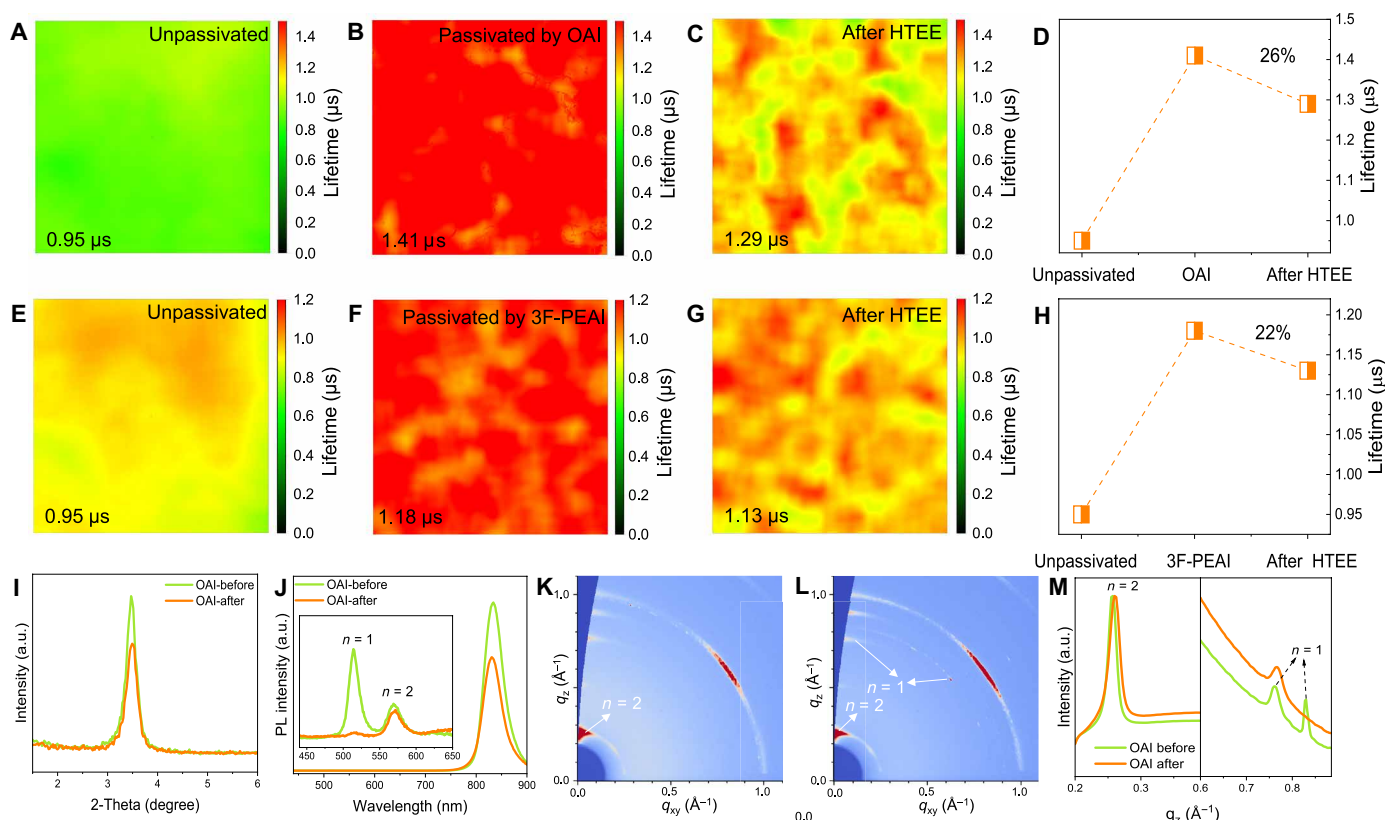


Fig. 3. Characterizations for rebound of defect density. (A to C) TRPL mapping of unpassivated, passivated by OAI and after HTEE perovskite films; 30 μm by 30 μm . (D) The average lifetimes of unpassivated, passivated by OAI, and after HTEE perovskite films. (E to G) TRPL mapping of unpassivated, passivated by 3F-PEAI, and after HTEE perovskite films; 30 μm by 30 μm . (H) The average lifetimes of unpassivated, passivated by 3F-PEAI, and after HTEE perovskite films. (I) Small-angle XRD pattern of OAI-passivated perovskite film before and after HTEE. (J) PL result of OAI-passivated perovskite film before and after HTEE. The inset is an amplified PL spectrum with wavelengths ranging from 440 to 650 nm. (K) GIWAXS result of OAI-passivated perovskite film. (L) GIWAXS result of OAI-passivated perovskite film after HTEE. (M) Integrated intensity of GIWAXS data along q_z for OAI-passivated film before and after HTEE.

flakes (marked as “GOG”; fig. S25), which meet the strict requirements mentioned above (42, 43). The GO as a hole transport material (44–47) also provides a more matched energy levels between underneath Spiro-OMeTAD and graphite to better extract holes (figs. S26 and S37). We also prepared GOG layers with different thicknesses (fig. S27), and the optimized thickness was 120 nm (figs. S28 and S36). We first found that GOG could ensure the XRD (fig. S29) and XPS (fig. S30) results of the perovskite films before and after HTEE consistent, indicating that it can effectively protect the perovskite film from the damage during HTEE. TOF-SIMS results further proved that all the components of perovskite films remained almost unchanged throughout the entire film before and after HTEE (Fig. 4, A to C), verifying that GOG can indeed eliminate the inconsistencies between the prepared film and the final film working in device.

We further designed PL characterization to integrate both the perovskite films without (marked as “control”) and with GOG (marked as “target”) into one sample to exclude batch differences and more rigorously verify the effect of GOG in eliminating the inconsistencies (note S6). It can be found that GOG enabled the PL intensity and TRPL lifetime of target perovskite film to maintain almost the same as those before HTEE (fig. S31). We also elaborately designed another experiment to simultaneously observe the control

and target samples in one field of view with identical instrument parameters by CLSM (note S7). The result demonstrated that GOG could effectively avoid the PL quenching on the perovskite films surface caused by additional defects during HTEE (fig. S32D), intuitively confirming that the inconsistencies between the film before and after HTEE was effectively eliminated.

Last, we prepared PSCs to further investigate the effect of eliminating inconsistencies at the device level. The unpassivated PSC delivered a power conversion efficiency (PCE) of 21.6% (fig. S33A). After surface passivating by OAI, both the control and target PSCs showed high PCE of 24.30 and 25.27% at forward scan, respectively (Fig. 4D). While the open-circuit voltage (V_{OC}) and fill factor (FF) were mainly weakened in the control sample. At the reverse scan, the elimination of inconsistencies also increased the V_{OC} from 1.16 to 1.18 V and FF from 81.50 to 83.17%, thus resulting in a PCE increase from 24.63 to 25.55% (Fig. 4E and fig. S34). The steady-state power output has also been increased from 23.9 to 25.1% (fig. S33B). The statistical results of 36 devices show that the elimination of inconsistencies mainly leads to the improvement of V_{OC} and FF (Fig. 4F and fig. S35), with highest FF of 85%. The improvement of V_{OC} is mainly contributed by two aspects: larger quasi-Fermi level splitting due to lower defects density on the film surface (48) and lower carrier interface recombination in the device due to more efficient interface extraction.

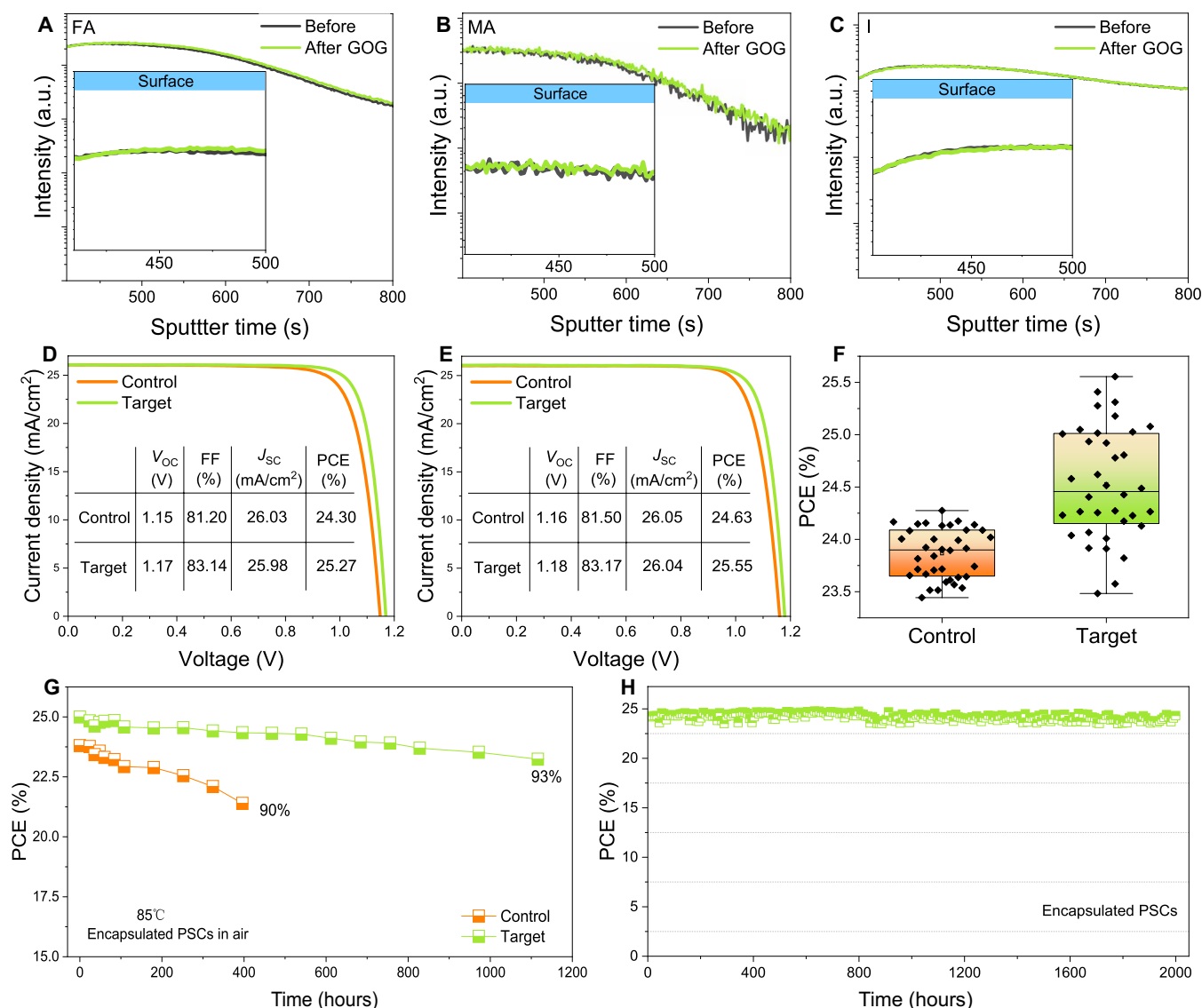


Fig. 4. Eliminating the inconsistencies. (A to C) TOF-SIMS results before and after HTEE with GOG. Inserts are enlarged surface regions. (D) *I*-*V* curves of control and target PSCs under forward scan. (E) *I*-*V* curves of control and target PSCs under reverse scan. (F) Statistical PCE of control and target PSCs. (G) The thermal stability of encapsulated PSCs under constant heating at 85°C in air (relative humidity = 20%). The Spiro-OMeTAD was substituted with PTAA {poly[bis(4-phenyl) (2,4,6-trimethylphenyl) amine]} in this characterization. (H) Maximum power point tracking result of encapsulated target PSC in N₂.

Stability, including internal ion migration and external resistance to light, heat, and water, is the most pressing issue for PSCs at present (49, 50). Among all the external invasions, moisture is most destructive to perovskite films. We validated that GOG can effectively insulate water from perovskite films (fig. S38A). Furthermore, we placed the integrated control and target half-film samples in a designed high-humidity environment while simultaneously recording their morphology changes over time (fig. S38B). After 60 hours, a large number of yellow phases have appeared in control half perovskite film, while the target half-film shows obviously stronger resistance to high moisture (fig. S38C). In addition, we also visually proved that GOG could effectively inhibited the migration of iodine on the basis of our delicately designed experiments (note S8 and movie S1). Because the decomposition reaction of perovskite films is reversible,

inhibiting the escape of perovskite components can greatly improve the film stability (33, 51, 52). We further placed the integrated control and target half-film samples under illumination of 1 sun at 85°C (fig. S40). The result validated that the target half-film sample has markedly higher stability against light/heat due to the effective suppression of component escape.

We also found that the gold (Au) from electrode in control PSCs would migrate to SnO₂ after operation at maximum power point (MPP) for 300 hours through TOF-SIMS (fig. S41), which was re-validated by an energy-dispersive spectrometer (fig. S42). The migration of Au was further demonstrated to reduce the conductivity of SnO₂ through device level *I*-*V* testing (fig. S43C). Moreover, we also found that the migration of Au changed the work function (W_F) of SnO₂ from -3.37 to -3.57 eV (fig. S43, D to F), thus increasing the

barrier of electron extraction at the buried interface by 0.2 eV. Fortunately, this unwanted Au migration was effectively inhibited in target PSCs (figs. S41 and S42). In general, downward migration of Au owing to operation and upward migration of FA-, MA-, and I-related ions because of HTEE damage and vacuum extraction could be effectively inhibited, maybe because of the smaller diameter of the staggered carbon atom ring compared to that of these ions. Therefore, it is difficult for them to migrate through the GOG layer (fig. S44). The violently enhanced film stability and the elimination of inconsistencies enable us to obtain highly stable PSCs. We first heated the encapsulated PSCs at 85°C in air with relative humidity ~ 20% to evaluate the thermal stability. The results showed that the target device could maintain 93% of its initial efficiency after 1100-hour aging (Fig. 4G and fig. S45), while the control PSC dropped to 90% within 400 hours. Operating stability was further investigated, and the target PSC exhibited negligible performance degradation after tracking at MPP for 2000 hours (Fig. 4H), among the most stable n-i-p PSCs (fig. S46 and table S1).

DISCUSSION

Ensuring that the perovskite film working in the device still retains the full excellent properties of the as-prepared film is critical to fully constructing high-performance PSCs. We found that HTEE will damage the perovskite film surface by facilitating the escape of organic and iodine components, thus introducing defects, disrupting interfacial energy levels, and deteriorating stability. Such neglected inconsistencies will unfortunately weaken the contribution of film optimization to the final device performance improvement. We designed a bilayer structure to effectively eliminate the undesired inconsistencies to save the loss of prefilm optimization and also markedly improved the stability of the perovskite film, intuitively proved by extensive delicately designed experiments. Therefore, we obtained highly efficient PSCs with a laboratory PCE of 25.55%, which maintained almost 100% of the original efficiency after operating for 2000 hours. We hope that our work could attract more attention on whether the film optimization is fully transformed to final device enhancement, including PSCs, light-emitting diodes, photodetectors, and other perovskite-based optoelectronic devices. We expect that our work could inspire thinking about whether the parameters of other soft lattice-based semiconductor materials are consistent with the material properties that actually work in the final device and whether the semiconductor parameters we tested in the past by the device can truly reflect the properties of the material.

MATERIALS AND METHODS

Materials

Indium tin oxide (ITO)/glass was purchased from Advanced Election Technology Co., Ltd. PbI₂ (superdry beads, 99.999%), CsCl (superdry beads, 99.999%), *N,N*-dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.9%), acetonitrile (99.9%), OAI and isopropyl alcohol (IPA, 99.7%), and chlorobenzene were purchased from Sigma-Aldrich. Natural graphite flake was purchased from Alfa Aesar. GO was purchased from Shanghai Yuanye Biotechnology Co., Ltd. 99.8% Lithium bis (trifluoromethanesulfonyl) imide (Li-TFSI, 98%), 4-tert-butylpyridine (96%), formamidinium iodide (FAI), methylammonium chloride (MACl), 2,2',7,7'-tetrakis (*N,N*-*p*-dimethoxyphenylamino)-9,9'-spirobifluorene (Spiro-OMeTAD, 99.8%) and SnO₂

colloid precursor [Tin (IV) oxide, 12% in H₂O colloidal dispersion] were purchased from Xi'an Polymer Light Technology Corporation. All the chemicals were used as received without further purification.

Solar cell device fabrication

Procedures previously described in our works were followed (5, 50). The high-transparent glasses/ITO ($\approx 15 \Omega \text{ sq}^{-1}$) were cleaned by ultrasonic cleaner for 35 min in deionized water, acetone, and IPA sequentially. Before use, the ITO-coated glass was dried by nitrogen gun and treated with 90-W air plasma for 60 s. Then, the SnO₂ solution (1:2.2 diluted by deionized water) was spin coated on the ITO at 3000 rpm for 30 s and annealed in ambient air at 150°C for 30 min. PbI₂ (691.5 mg) in DMF:DMSO = 0.9 ml:0.1 ml was stirred at 70°C for an hour and filtered with a 0.22- μm polytetrafluoroethylene (PTFE) filter. After the films were cooled down to room temperature, 60 μl of PbI₂ solution was spin coated on SnO₂ at 1800 rpm for 30 s and then annealed at 70°C for 1 min. Subsequently, 100 μl of filtered organic salt solution containing FAI:MAI:MACl (90 mg:6 mg:9 mg in 1 ml of IPA) was spin coated on the room-temperature PbI₂ films at 2000 rpm for 30 s. Then, the films should be transferred from glovebox to the hot plate in the atmosphere (25 to 35% humidity) for annealing at 150°C for 15 min (5, 50). After that, 8 mg of OAI was dissolved in 1 ml of IPA, after filtering by a 0.22- μm PTFE filter, this solution was spin coated onto the upper surface of the perovskite film at 5000 rpm for 30 s. Subsequently, the film was annealed at 100°C for 6 min. Subsequently, mixed Spiro-OMeTAD solution was prepared by mixing 433.8 mg of spiro-OMeTAD in 6 ml of chlorobenzene, 172 μl of a 4-tert-butylpyridine solution, and 105 μl of Li-TFSI solution (520 mg ml⁻¹ in acetonitrile), respectively, and was spin casted on the surface of the perovskite films at 6000 rpm for 30 s. After that, the films need to be oxidized for 12 hours in a drying cabinet with a relative humidity of 1%. For the target sample, GO dispersed in IPA (1 mg/ml) was sonicated for 30 min and then filtered by a 0.22- μm PTFE filter, and then this solution was spin coated on spiro-OMeTAD at 5000 rpm for 30 s. Subsequently, natural graphite flakes were finely coated onto the surface of the film by cotton swabs or soft brushes (42). The film was gently cleaned by a nitrogen gun. Last, gold electrode was thermally evaporated with the thickness of about 90 nm. The evaporation speed was 0.01 nm/s at the first 15 nm and 0.03 nm/s for the rest of the procedure. Then, the PSCs should be oxidized for 30 hours in a drying cabinet with a relative humidity of 1% before characterization (5, 50).

Perovskite film characterization

The crystal structure and phase composition were determined by the XPert Pro (Holland) x-ray diffract meter with Cu Ka1 radiation ($\lambda = 0.154 \text{ nm}$). The surface electronic states were investigated by the ultraviolet (UV) photoelectron spectroscopy (Axis Ultra, Kratos Analytical Ltd) and the nonmonochromated He I _{α} ($h\nu = 21.22 \text{ eV}$) was the photon source. The absorption spectra of the perovskite layers were measured by UV-visible spectrometry using a UV-2600 (Shimadzu). The PL and TRPL were investigated at room temperature using FLS1000 (Edinburgh Instruments) spectrometer, and the laser fluence is 35 nJ/cm² (5, 50). The CLSM characterization were performed on homemade Olympus FV3000 CLSM system with 60 $\times$ objective lens. The surface and cross-sectional morphology of the perovskite films and device were recorded by a scanning electron microscope (SEM) (Quanta FEG 450) under 10-kV electron beam accelerate voltage. The spatial distribution of the AFM-IR spectroscopy

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